

Frank Shan

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HIGHLIGHTS

- **Full-stack** developer specialized in translating **business requirements into production-ready** systems. Shipped a complete product in under 3 months. Expert in **building MVP** rapidly using coding agents.

EDUCATION

University of Waterloo | *CGPA: 92.3/100* Sep 2024 - Apr 2029
Bachelor of Computer Science, Honours Computer Science, AI Specialization

EXPERIENCE

Software Developer Intern | *Richmond, BC* Sep 2025 - Apr 2026
Ziyutec Inc.

- **Owned** the end-to-end planning and development of a [global B2B marketplace](#) for a startup initiative, **leading** system design, implementation, and deployment (AWS) across the **entire SDLC**.
- Architected a Role-Based Access Control framework (NestJS, PostgreSQL) to support extensible user roles; balanced database **normalization** against **query performance** via denormalization and data **integrity** checks.
- Engineered a **caching** layer to decouple client-side state from **atomic** database transactions, reducing **user friction** while ensuring transactional integrity.
- Built an automation agent for engineering workflow, reducing operational workload by **20%** across a **50-person** engineering team.
- Built a **RAG agent**, indexing **500+** **enterprise documents** of different format; improved **contextual relevance by 16%** (DeepEval) via **hybrid search and reranking** on synthetically generated evaluation set.
- Maintained and enhanced a **multi-tenant** ERP platform comprising **60K+** **lines of legacy code**, supporting **1,000+** **active projects**. Resolved a **long-standing** session management defect by implementing **JWT refresh token rotation**, eliminating form data loss.

Research Assistant | *Waterloo, ON* Feb 2026 - Present
University of Waterloo

- Designed comparative experiments for a probabilistic programming language PAL that compiles structural rules into tensor operations. Built a **hybrid neuro-symbolic** framework for training Named Entity Recognition models to enforce strict linguistic constraints and improve early-epoch model convergence.

PROJECTS

Compact Vision-Language Model | *PyTorch, Transformers* [github](#) 

- Built a LLaVA style **Vision-Language Model**, fusing a **SigLIP-2** vision encoder with **Qwen3-0.6B** via a pixel-shuffle modality projector; achieved **32.9% zero-shot accuracy on MMStar**.
- Designed a high-throughput training pipeline with a **greedy knapsack** batching strategy to maximize token-level GPU utilization, enabling efficient multimodal pretraining at **consumer-scale hardware**.

High Throughput LLM Inference Engine | *PyTorch* [github](#) 

- Built a LLM **inference engine from scratch**, featuring continuous batching, scheduler-driven execution, and pluggable dense/paged KV-cache backends, with **prefix caching** for efficient memory reuse.
- Designed a ShareGPT-based **benchmarking** pipeline and demonstrated up to **59% higher request throughput** and significantly lower TTFT with paged attention.

SKILLS

Programming Languages: `built-in method items of dict object at 0x1069ea1c0`

Frameworks: `built-in method items of dict object at 0x1069f8f80`

Tools: `built-in method items of dict object at 0x1069e9a80`

PROFESSIONAL CERTIFICATES

XCS224N, Stanford Online
Natural Language Processing with Deep Learning

December 2025
[Certificate of Achievement](#) 